Reciprocal inhibition entrains detuned PING circuits

exp078 · 5 August 2026 · Draft

Abstract

Reciprocal inhibition entrained two independently driven, slightly detuned PING circuits without a shared input or simulator change. The uncoupled pair oscillated at 42.725 Hz and 39.063 Hz, a 8.96% separation, with phase-locking value 0.097 and 3.662 Hz residual frequency difference. Reciprocal GABA coupling at strength 0.2 and 0.1 ms reduced that difference to 0 Hz, raised phase locking to 0.818, and raised mean 30–80 Hz coherence from 0.465 to 0.612. Four registered cells satisfy all locking gates with an adjacent locked neighbour. Strong short/intermediate coupling instead silenced circuit B. This one-seed result demonstrates the mechanism and its suppression boundary; it does not estimate generality.

Registered goal

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/goal Design and execute exp078 as the first graph-native gamma-coupling
experiment, building directly on the two-circuit executor validated by exp077
and ar063 Milestone 3.

Outcome:
Demonstrate whether reciprocal inhibitory coupling can entrain two
independently driven, slightly detuned PING circuits. Establish a
scientifically interpretable transition from uncoupled phase drift, through
stable frequency/phase locking, to excessive-coupling suppression where
present.

Repository and publication:
- Begin from current main after confirming the disposition of PR #64.
- If PR #64 has not been merged, stop and ask before merging it or building
dependent work on another base.
- Create a focused autoresearch branch and draft PR for exp078.
- Do not merge the new PR.
- Preserve exp077 unchanged as the Milestone 3 architecture/causality record.
- Update ar063 Milestone 4 only after its registered acceptance gates pass.

Scientific design:
- One exploratory seed only.
- Run locally unless paid compute is later explicitly authorized.
- Use two independently named PING circuits constructed through snnlang and
executed by the graph-native tools/snn executor.
- Use approximately 40 excitatory and 10 inhibitory neurons per circuit,
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unless a nearby size is materially better supported by existing reusable
components.
- Give the circuits independent stochastic or tonic spike drives. Do not use
shared input.
- Slightly detune their uncoupled natural gamma frequencies.
- Simulate long enough to resolve phase and frequency reliably: target 1.5–
2.0 seconds, discarding the first 300–500 ms as transient.
- Keep initial state and input realizations matched across coupling variants.
- Couple each inhibitory population to the other circuit’s excitatory
population with reciprocal GABA projections.
- All sweep variants must differ only through graph data; simulator-specific
edits invalidate the experiment.

Staged protocol:

1. Audit and registration
- Read ar063, exp077, its artifacts, and the current snnlang/tools/snn
interfaces.
- Record the exact hypothesis, calibration procedure, sweep grid, activity
criteria, synchronization criteria, suppression criteria, success criterion,
and kill criterion before interpreting coupling results.
- Do not change these criteria after seeing the registered sweep.

2. Local smoke
- Verify that two small uncoupled circuits execute, remain finite, expose
named E/I recordings, and produce reproducible outputs.
- Verify exact delay timing and that coupling variants require graph changes
only.

3. Bounded uncoupled calibration
- Calibrate only input drive and, if necessary, existing within-circuit E↔I
strengths.
- Select two uncoupled circuits that:
  - remain finite and active after the transient;
  - have neither near-zero nor saturated E/I activity;
  - each show a clear population-rate peak in approximately the 30–80 Hz
band;
  - differ in dominant frequency by approximately 5–15%.
- Pre-register a bounded candidate grid and deterministic selection rule
before running it.
- Do not tune cross-coupling during this stage.
- Once selected, freeze all within-circuit and input settings for the
coupling sweep.
- If no candidate pair satisfies the criteria, record the failure and stop
rather than silently expanding the search.

4. Registered coupling sweep
- Include the uncoupled baseline.
- Sweep a compact set of approximately 4–5 reciprocal inhibitory coupling
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strengths, spanning ineffective through near-suppressive coupling.
- Sweep three biologically interpretable delays: short, intermediate, and
approximately half of the uncoupled gamma period.
- Target roughly 13–16 total graph variants.
- Use the identical seed, input spike realizations, initial conditions,
simulation duration, analysis window, neuron counts, and within-circuit
parameters for every condition.
- Do not add common drive, train weights, or introduce another
synchronization mechanism.

Required measurements:
- E and I spike rasters for both circuits.
- Smoothed population firing rates.
- Mean E and I firing rates and active-cell fractions.
- Dominant gamma frequency and peak prominence for each circuit.
- Absolute frequency difference.
- Phase-locking value using a declared band/filter and analysis window.
- Mean circular phase difference.
- Coherence around the gamma peak.
- Cross-correlation peak and lag.
- Explicit silent, saturated, non-finite, or spectrally invalid condition
flags.
- Runtime and peak memory.
- Graph digests, canonical diagrams, manifests, delay settings, recordings,
provenance, and reproducer for every variant.

Interpretation criteria:
- “Active” requires both circuits to retain valid E and I activity after the
transient.
- “Locked” requires, at minimum:
  - both circuits active;
  - substantially reduced frequency difference relative to uncoupled;
  - increased phase-locking value and gamma-band coherence relative to
uncoupled;
  - a stable, reportable phase offset rather than transient coincidence.
- Pre-register numerical thresholds using the uncoupled calibration evidence
before running the coupling sweep.
- Do not call suppression synchronization.
- Treat isolated single-grid-cell improvements cautiously; prefer a
contiguous strength/delay region.
- Examine whether phase lag varies systematically with delay.
- Make no population-level or generality claim from one seed. This is an
exploratory mechanism demonstration.

Success criterion:
At least one pre-registered moderate-coupling condition keeps both circuits
active and satisfies the locking thresholds, while the uncoupled baseline
exhibits the registered detuning and weaker phase locking. The result must be
supported jointly by rasters, population rates, frequency difference, PLV,
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coherence, and phase-lag evidence.

Kill criterion:
Stop and record a negative result if:
- suitable active, detuned uncoupled oscillators cannot be obtained within
the registered calibration grid;
- coupling only suppresses activity or produces invalid/saturated dynamics;
- no registered coupling condition meets the locking thresholds;
- synchronization requires shared input, simulator-specific edits,
unregistered tuning, or changing the scientific criteria;
- or completion requires paid compute or authority not already granted.

Implementation constraints:
- Prefer implementing exp078 through experiments/exp078.py and existing
snnlang/tools/snn capabilities.
- Existing authorization to edit tools/snn does not carry into this fresh
goal unless explicitly included by the user. Stop and ask before editing
tools/snn.
- Do not modify historical experiment runners merely to adopt the new API.
- Do not run the full repository test suite on the host.
- Use focused unit, schema, causality, recording, numerical, and artifact
checks plus full repository lint/type checks where lightweight.
- Record meaningful failed or killed attempts when they affect
interpretation.
- Do not use paid compute without explicit authorization.

Artifacts and writing:
- Produce artifacts/data/exp078 containing:
  - numbers.json;
  - provenance and exact reproducer;
  - exact goal prompt;
  - timestamped activity log;
  - calibration candidates and deterministic selection record;
  - complete registered sweep table;
  - graph bundles, manifests, digests, and canonical diagrams;
  - input spike realizations;
  - named E/I/voltage/conductance recordings;
  - firing-rate and spectral diagnostics;
  - PLV, phase, coherence, and cross-correlation measurements;
  - matched rasters and population-rate figures;
  - strength × delay heatmaps;
  - runtime and memory measurements.
- Write writings/exp078.typ as a cold-readable report.
- Put the full goal prompt immediately below the abstract.
- Include the registered design, calibration outcome, exclusions, failures,
limitations, and timestamped activity-log appendix.
- Every reported number and figure must come from committed artifacts.
- If the experiment passes, update ar063 B.7 Milestone 4 with links to the
evidence and relevant commits.
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- If it fails, preserve Milestone 4 as incomplete and document exactly why.

Git and review:
- Use focused commits for registration, implementation, calibration,
registered execution, evidence, and writing.
- Build Demolab successfully.
- Visually inspect every page of the rendered exp078 and updated ar063 PDFs.
- Push the branch and open a draft PR against main.
- Add timestamped PR comments at meaningful calibration, execution, failure,
and validation milestones.
- Do not merge.

Finish at the human-review gate with:
- whether coupling-induced synchronization was demonstrated;
- the exact calibrated frequencies and activity diagnostics;
- the winning or negative strength/delay results;
- frequency difference, PLV, coherence, phase lag, firing rates, and
suppression evidence;
- validation performed;
- known limitations and remaining work;
- exact compute spend;
- links to rendered exp078/ar063, artifacts, commits, and the draft PR.
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Design registered before coupling

The hypothesis was that moderate reciprocal I→E GABA coupling would entrain two active, independently driven PING circuits while zero coupling permitted drift and excessive coupling could suppress activity. Both circuits contain 40 excitatory and 10 inhibitory conductance neurons. Independent Poisson generators drive 16 channels per circuit; their seeds are archived separately, so no common-input synchrony is possible. Every condition uses seed 78, a 0.1 ms timestep, 1800 ms duration, and discards the first 400 ms.

The bounded calibration changed input rate and input weight only. The reusable snnlang PING component fixed all within-circuit E↔I settings. A candidate had to retain finite, non-silent, non-saturated E/I activity, a prominent 30–80 Hz rate peak in each circuit, and 5–15% detuning. The deterministic score selected the valid candidate nearest 10% detuning, with peak prominence and grid order breaking ties. Cross-coupling was not evaluated during calibration.

Locking required both circuits to remain active, frequency difference no more than 50% of baseline, phase-locking gain at least 0.15, mean-band coherence gain at least 0.1, phase-locking value at least 0.55 in each half-window, and phase-offset drift no more than 0.6 rad. A passing cell also needed an orthogonally adjacent locked cell. Silent conditions were never eligible.

Calibration produced active detuned gamma

Diagnostic	Circuit A	Circuit B
Dominant frequency	42.725 Hz	39.063 Hz

Peak prominence	63	199.3
Excitatory rate	37.5 Hz	34.839 Hz
Inhibitory rate	120.429 Hz	110.5 Hz
E active-cell fraction	1	1
I active-cell fraction	1	1

Candidate 8 used input weight 0.4 and independent per-channel drive rates 260 Hz and 210 Hz. All cells participated. The first analysis implementation took the maximum coherence anywhere in the gamma band; its uncoupled value was 0.992, making a +0.10 gain impossible. Before any coupling condition ran, the declared coherence statistic was frozen as the arithmetic mean of magnitude-squared coherence bins from 30 through 80 Hz. The corrected uncoupled baseline is 0.465.

A contiguous locking region emerged

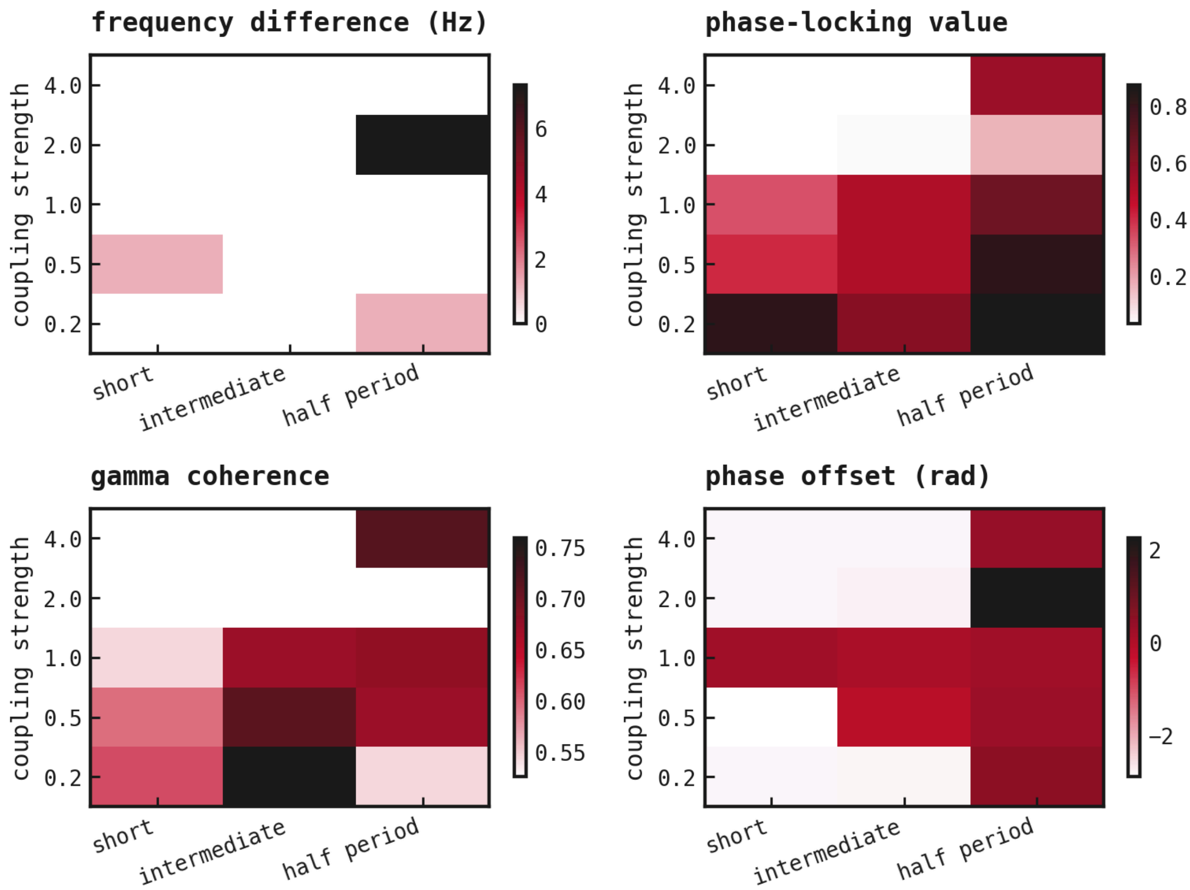

Figure 1: The complete registered strength × delay sweep. Low-to-moderate coupling removes the frequency difference and increases phase locking and coherence. Delay changes the stable phase offset. Missing spectral values correspond to explicitly flagged suppression, not synchronization.

Condition	Δf	PLV	Coherence	Phase	Lag
Uncoupled	3.662 Hz	0.097	0.465	−2.679	34.2 ms
0.2, short	0 Hz	0.818	0.612	−2.826	−17.5 ms

0.2, intermediate	0 Hz	0.598	0.759	−2.806	−18 ms
0.5, half period	0 Hz	0.817	0.67	0.29	32.2 ms

The four locked cells are 0.2/short, 0.2/intermediate, 0.5/half-period, and 1.0/half-period. Each has an orthogonally adjacent locked neighbour, so the result is not an isolated best cell. Short and intermediate delay at strength 0.2 settle near an anti-phase offset of −2.826 and −2.806 rad. Half-period delay changes the locked offset to 0.29 rad and the cross-correlation lag to 32.2 ms. The sign and size of phase/lag therefore change systematically with the delay regime.

Rasters and rates distinguish locking from silence

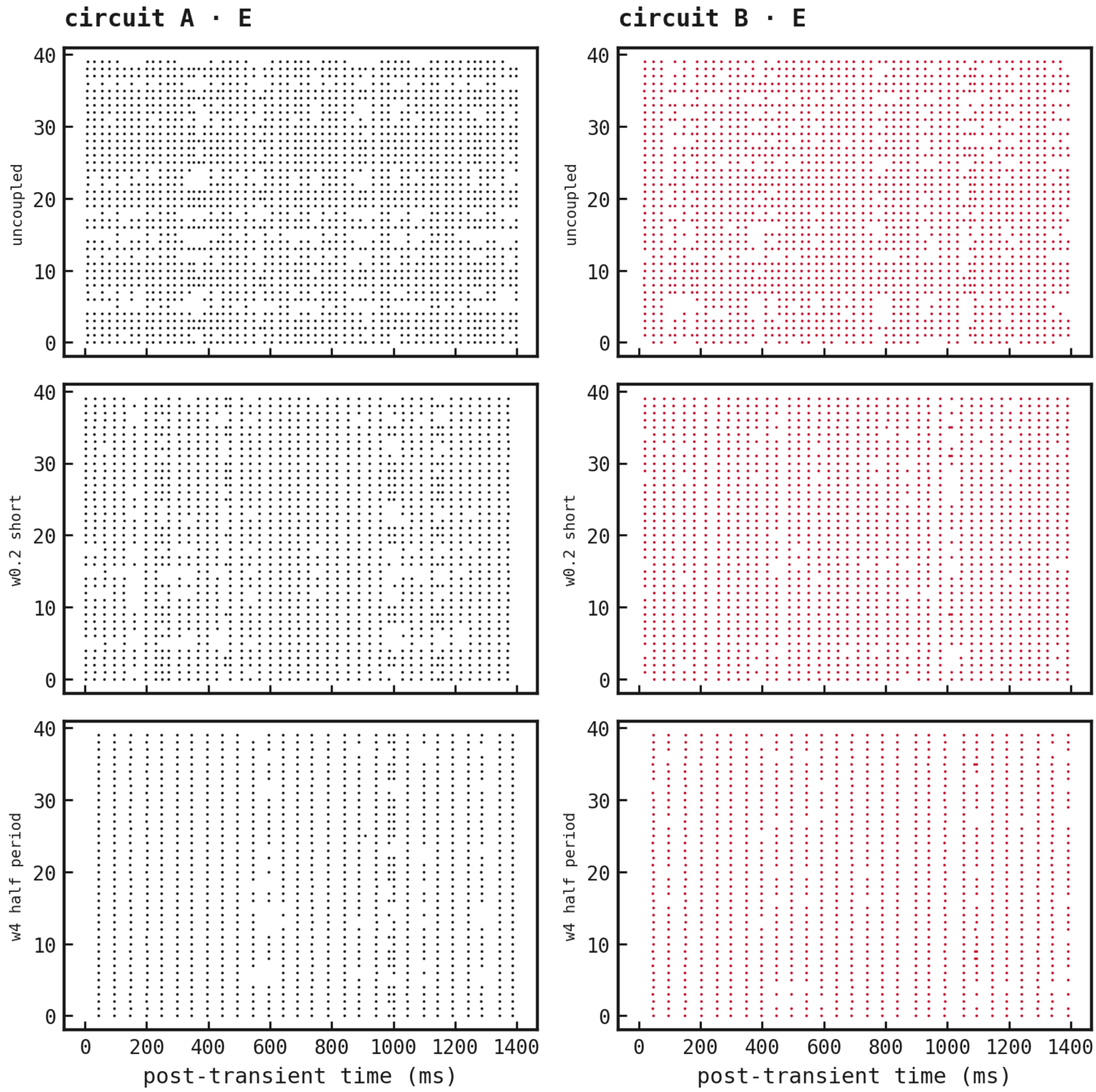

Figure 2: Matched excitatory rasters after the transient for the uncoupled baseline, the first registered locked condition, and the strongest half-period condition. Inputs, seed, initial state, and within-circuit parameters are identical across rows.

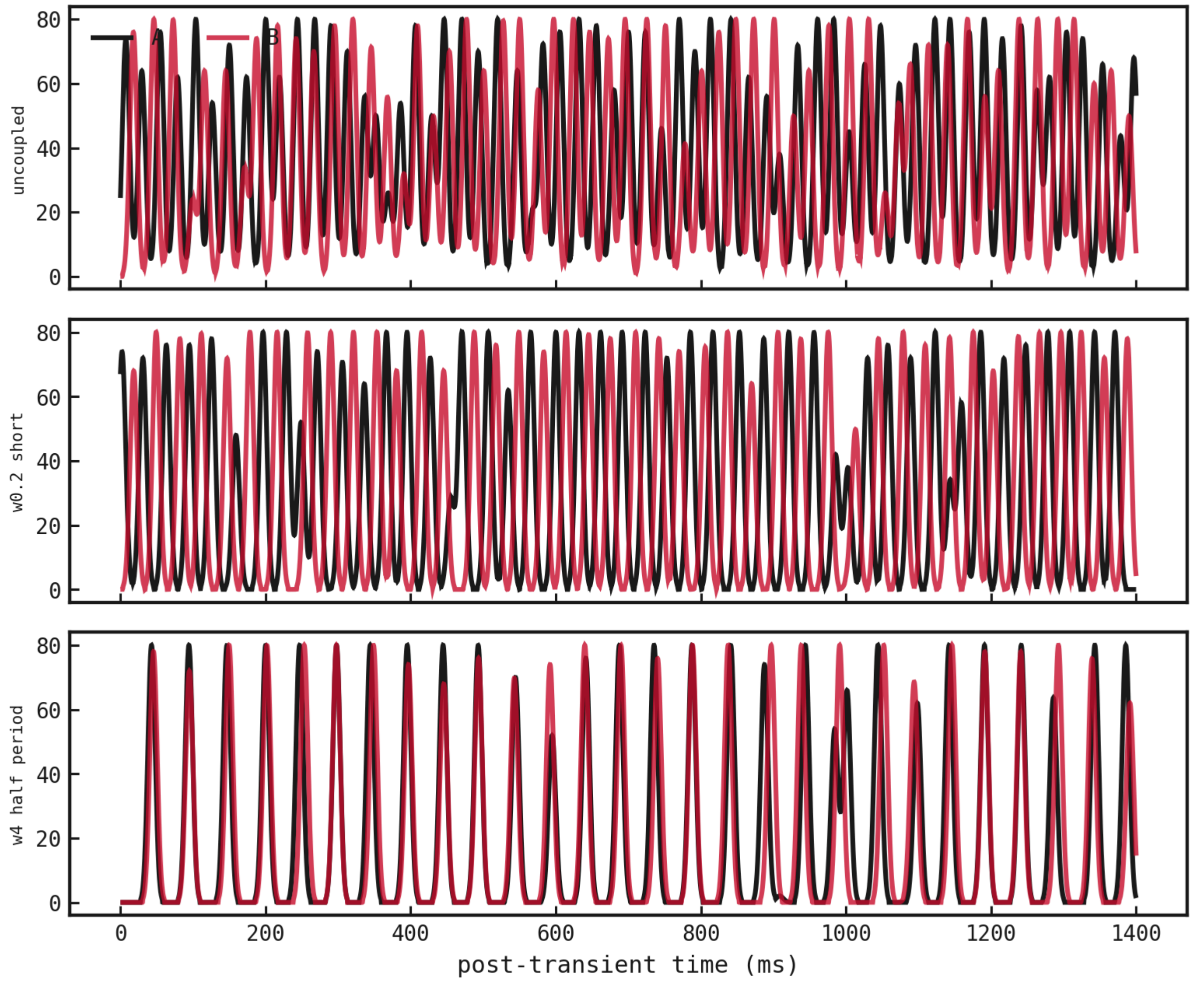

Figure 3: Smoothed excitatory population rates for the same conditions. The coupled pair shares a rhythm with a stable offset; the baseline drifts.

At strength 2 and short delay, circuit B has 0 Hz excitatory and 0 Hz inhibitory activity, while circuit A remains at 37.5 Hz and 120.429 Hz. The condition is explicitly inactive and spectrally invalid. Strength 2/intermediate and strengths 4 at short/intermediate delay show the same one-sided suppression boundary. The half-period delay avoids complete silence even at the largest strengths, but those cells fail the full locking gates.

Graph-native execution and evidence

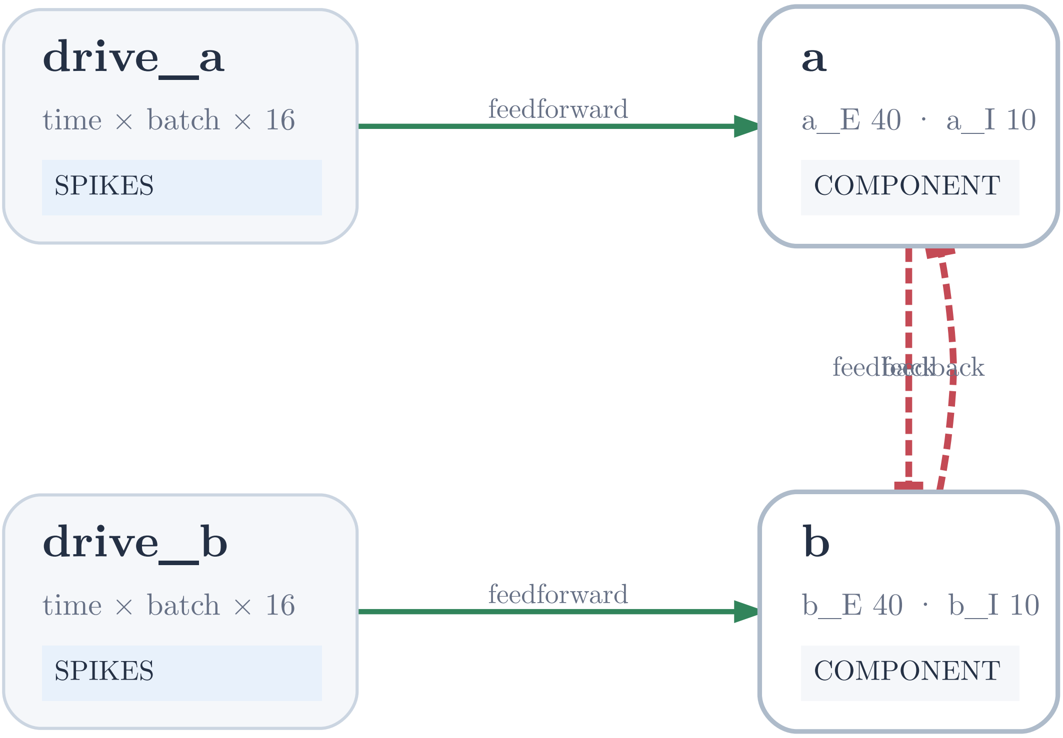

Figure 4: Representative reciprocal graph. Each inhibitory population sends a delayed GABA projection to the other circuit's excitatory population; the two input nodes are independent.

All 16 conditions differ only through graph data. Each archives its graph, authenticated manifest, canonical diagrams, graph digest, raw named E/I, voltage, and conductance recordings, firing-rate and spectral traces, runtime, and peak traced Python memory. The complete registered sweep took 18m 32s locally. Simulator edits were 0 and paid compute cost was $0.

The first complete sweep attempt computed all conditions but failed atomically before publication: derived rate arrays had overwritten raw spike keys in the figure map. Commit `a781c7f` records that killed attempt and an experiment-side key fix. The unchanged rerun published the evidence in commit `03bfbe9`. `tools/snn`, exp077, the registered grid, input realization, metrics, and gates were unchanged.

Conclusion and limitations

The registered success criterion passes. Reciprocal inhibition alone moves this pair from independent phase drift into a contiguous active locking region, and excessive short/intermediate coupling produces suppression rather than a misleading synchrony score. This supports graph-native Milestone 4 as a mechanism demonstration and shows that scientifically meaningful variants can be authored in graph data without simulator edits.

The evidence has one exploratory seed, one calibrated oscillator pair, dense connectivity, and one set of intrinsic PING parameters. It does not estimate robustness across seeds, circuit sizes, detuning ranges, input statistics, or biological preparations. The run manifest is worktree-dirty only because a concurrent Demolab preview regenerated tracked PDFs; its code-specific dirty flag is false and no patch affected execution.

Timestamped activity log

2026-08-05T12:48:57Z

Registered the hypothesis, bounded calibration, deterministic selection, graph-only sweep, locking thresholds, success criterion, and kill criterion before coupling execution.

2026-08-05T12:50:00Z

Passed the local smoke gate with finite named E/I and voltage recordings, exact replay, graph-only coupling, and exact five-step delay lowering.

2026-08-05T13:02:03Z

Completed the bounded 12-candidate uncoupled calibration locally and selected candidate 8 deterministically; no cross-coupling was evaluated.

2026-08-05T13:03:12Z

Replaced maximum-over-band coherence with registered 30–80 Hz mean coherence before coupling because the uncoupled maximum saturated at 0.992 and made the declared gain impossible.

2026-08-05T13:14:54Z

Froze the selected uncoupled baseline and all numerical thresholds before the first coupling condition.

2026-08-05T13:34:17Z

Killed the first complete sweep publication after rate traces overwrote spike-array artifact keys and raster rendering failed; atomic staging prevented invalid evidence publication.

2026-08-05T13:54:37Z

Completed the unchanged pre-registered local coupling sweep after an experiment-side artifact-key fix, without simulator edits or paid compute.